

● MEDIUM

● ALGEBRA

● ANSWER KEY

Linear functions

05

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Q1**39**

The correct answer is 39. It's given that for the linear function j , m is a constant and $j12 = 18$. Substituting 12 for x and 18 for jx in the given equation yields $18 = m12 + 144$. Subtracting 144 from both sides of this equation yields $-126 = m12$. Dividing both sides of this equation by 12 yields $-10.5 = m$. Substituting -10.5 for m in the given equation, $jx = mx + 144$, yields $jx = -10.5x + 144$. Substituting 10 for x in this equation yields $j10 = -10.5(10) + 144$, or $j10 = 39$. Therefore, the value of $j10$ is 39.

Q2**241**

The correct answer is 241. For a certain animal, it's given that a model predicts the animal weighed 241 pounds when it was born and gained 3 pounds per day in its first year of life. It's also given that this model is defined by an equation in the form $fx = a + bx$, where fx is the predicted weight, in pounds, of the animal x days after it was born, and a and b are constants. It follows that a represents the predicted weight, in pounds, of the animal when it was born and b represents the predicted rate of weight gain, in pounds per day, in its first year of life. Thus, the value of a is 241.

Q3**C**

C. $f(x) = -\frac{4}{3}x + \frac{1}{3}$

Choice C is correct. A linear function can be written in the form $f(x) = mx + b$, where m is the slope and b is the y-coordinate of the y-intercept of the line. The slope of the graph can be found using the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. Substituting the values of the given points into this formula yields $m = \frac{-5 - 3}{4 - (-2)}$ or $m = \frac{-8}{6}$, which simplifies to $m = -\frac{4}{3}$. Only choice C shows an equation with this slope.

Choices A, B, and D are incorrect and may result from computation errors or misinterpreting the given information.

Q4**B**

B. 7

Choice B is correct. A linear function has a constant rate of change, and any two rows of the table shown can be used to calculate this rate. From the first row to the second, the value of x is increased by 2 and the value of $f(x)$ is increased by $6 = 4 - (-2)$. So the values of $f(x)$ increase by 3 for every increase by 1 in the value of x . Since $f(2) = 4$, it follows that $f(2+1) = 4 + 3 = 7$. Therefore, $f(3) = 7$.

Choice A is incorrect. This is the third x -value in the table, not $f(3)$. Choices C and D are incorrect and may result from errors when calculating the function's rate of change.

Q5**2850**

The correct answer is 2,850. It's given that the function $fx = 45x + 600$ gives the monthly fee, in dollars, a facility charges to keep x crates in storage. Substituting 50 for x in this function yields $f50 = 4550 + 600$, or $f50 = 2,850$. Therefore, the monthly fee, in dollars, the facility charges to keep 50 crates in storage is 2,850.

Q6**B**

B. 122

Choice B is correct. It's given that $fx = 2x + 244$ represents the perimeter, in centimeters (cm), of a rectangle with a length of x cm and a fixed width. If w represents a fixed width, in cm, then the perimeter, in cm, of a rectangle with a length of x cm and a fixed width of w cm can be given by the function $fx = 2x + 2w$. Therefore, $2x + 2w = 2x + 244$. Subtracting $2x$ from both sides of this equation yields $2w = 244$. Dividing both sides of this equation by 2 yields $w = 122$. Therefore, the width, in cm, of the rectangle is 122.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q7**D**

D. $B = 212 - \frac{x}{550}$

Choice D is correct. It's given that the boiling point of water at sea level is 212°F and that for every 550 feet above sea level, the boiling point of water is lowered by about 1°F. Therefore, the change in the boiling point of water x feet above sea level is represented by the expression

$-\frac{x}{550}$. Adding this expression to the boiling point of water at sea level gives the equation for

the boiling point B of water, in °F, x feet above sea level: $B = -\frac{x}{550} + 212$, or

$$B = 212 - \frac{x}{550}.$$

Choices A and B are incorrect and may result from using the boiling point of water at sea level as the rate of change and the rate of change as the initial boiling point of water at sea level.

Choice C is incorrect and may result from representing the change in the boiling point of water as an increase rather than a decrease.

Q8**67**

The correct answer is 67. It's given that $f5 = 32$. Therefore, for the given function f , when $x = 5$, $fx = 32$. Substituting 5 for x and 32 for fx in the given function $fx = 4x + kx - 1$ yields $32 = 4(5) + k(5) - 1$, or $32 = 20 + 5k$. Subtracting 20 from each side of this equation yields $12 = 5k$. Dividing each side of this equation by 5 yields $k = \frac{12}{5}$. Substituting $\frac{12}{5}$ for k in the given function $fx = 4x + kx - 1$ yields $fx = 4x + \frac{12}{5}x - 1$, which is equivalent to $fx = 4x + \frac{12}{5}x - 1$, or $fx = \frac{32}{5}x - 1$. Substituting 10 for x into this equation yields $f10 = \frac{32}{5}(10) - 1$, or $f10 = 67$. Therefore, the value of $f10$ is 67.

Q9**D**

D. 12,0

Choice D is correct. The given function f is a linear function. Therefore, the graph of $y = fx$ in the xy -plane has one x -intercept at the point $k, 0$, where k is a constant. Substituting 0 for fx and k for x in the given function yields $0 = 7k - 84$. Adding 84 to both sides of this equation yields $84 = 7k$. Dividing both sides of this equation by 7 yields $12 = k$. Therefore, the x -intercept of the graph of $y = fx$ in the xy -plane is 12,0.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q10**2**

The correct answer is 2.6. According to the model, the head width, in millimeters, of a worker bumblebee can be estimated by adding 0.6 to 4 times the body weight, in grams, of the bee. Let x represent the body weight, in grams, of a worker bumblebee and let y represent the head width, in millimeters. Translating the verbal description of the model into an equation yields $y = 0.6 + 4x$. Substituting 0.5 grams for x in this equation yields $y = 0.6 + 4(0.5)$, or $y = 2.6$. Therefore, a worker bumblebee with a body weight of 0.5 grams has an estimated head width of 2.6 millimeters. Note that 2.6 and $13/5$ are examples of ways to enter a correct answer.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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